DESIGN & IMPLEMENTATION OF A DATABASE MANAGEMENT SYSTEM

Scholarship & Financial Aid Management

REVIEW 2

Database Design, Normalization & SQL Implementation

OCTOBER 13, 2026



Review 2 Objectives

From logical design to a working, queryable prototype.

01

Normalized design up to 3NF

02

Functional dependencies

03

Data dictionary development

04

DDL + integrity constraints

05

Sample data population

06

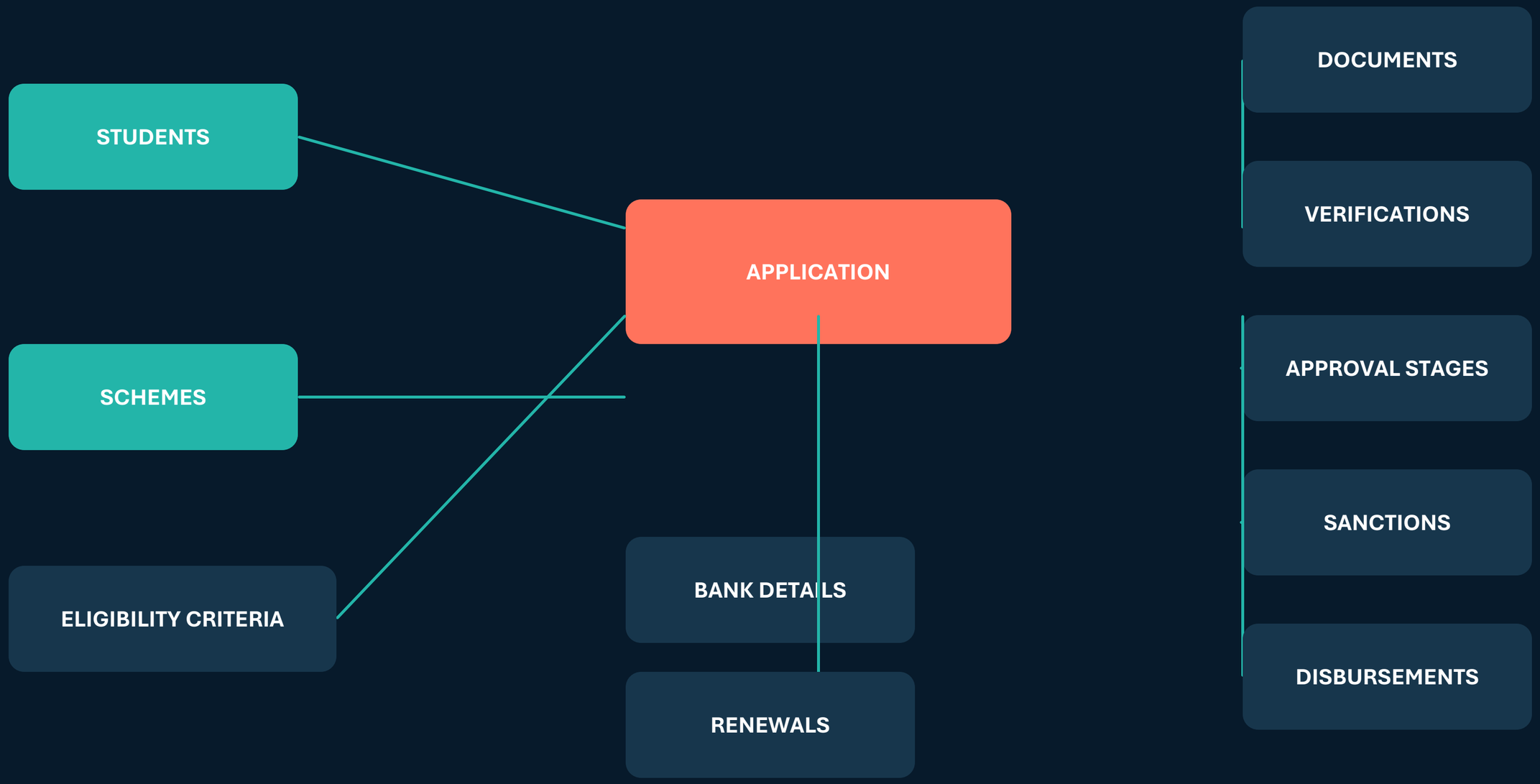
SQL queries + views

07

Prototype demonstration

Final ER Model

A connected model for the complete scholarship lifecycle



Relational Schema — Part 1

Core entities • primary keys in coral • foreign keys in teal

STUDENT

Student_ID PK

Name · DOB · Program
Income_Category · CGPA

SCHEME

Scheme_ID PK

Scheme_Name · Type · Amount

APPLICATION

Application_ID PK

Student_ID FK · Scheme_ID FK
Year · Status · Submit_Date

DOCUMENT

Document_ID PK

Application_ID FK
Document_Type · Status

BANK_DETAILS

Bank_ID PK

Student_ID FK
Account_No · IFSC

Relational Schema — Part 2

Lifecycle entities extend the application record without duplicating it.

ELIGIBILITY_CRITERIA	`Criteria_ID PK`	Scheme_ID FK · Criterion · Requirement
VERIFICATION	`Verification_ID PK`	Application_ID FK · Document_ID FK · Status
APPROVAL_STAGE	`Approval_ID PK`	Application_ID FK · Stage · Status · Approval_Date
SANCTION	`Sanction_ID PK`	Application_ID FK · Amount · Sanction_Date
DISBURSEMENT	`Disbursement_ID PK`	Application_ID FK · Amount · Date · Mode
RENEWAL	`Renewal_ID PK`	Application_ID FK · Renewal_Year · Status

Functional Dependencies

Each identifier determines one complete, stable fact set.

```
Student_ID → Name, DOB, Program, Income_Category, CGPA

Scheme_ID → Scheme_Name, Type, Amount

Application_ID → Student_ID, Scheme_ID, Year, Status, Submit_Date

Document_ID → Application_ID, Document_Type, Status

Sanction_ID → Application_ID, Amount, Sanction_Date

Disbursement_ID → Application_ID, Amount, Date, Mode

Renewal_ID → Application_ID, Renewal_Year, Status
```

Normalization up to 3NF

A deliberate progression from clean fields to clean dependencies.

1NF

Atomic values

No repeating groups
Single-valued fields

2NF

Full dependency

1NF satisfied
Depends on the entire key

3NF

No transitivity

2NF satisfied
Separation of concerns

RESULT Reduced redundancy • stronger integrity • optimized design

Data Dictionary

A compact reference for keys and business attributes.

TABLE	PRIMARY KEY	KEY ATTRIBUTES
Student	`Student_ID`	Name, DOB, Program, CGPA
Scheme	`Scheme_ID`	Name, Type, Amount
Application	`Application_ID`	Student_ID, Scheme_ID, Status
Document	`Document_ID`	Application_ID, Type, Status
Verification	`Verification_ID`	Application_ID, Status
Sanction	`Sanction_ID`	Application_ID, Amount
Disbursement	`Disbursement_ID`	Application_ID, Amount, Date
Renewal	`Renewal_ID`	Application_ID, Year, Status

DDL & Integrity Constraints

Integrity is enforced in the database—not left to chance.

PRIMARY KEY
Unique identity

FOREIGN KEY
Referential integrity

UNIQUE
No duplicate values

NOT NULL
Required fields

CHECK
Valid value ranges

DEFAULT
Safe fallback values

```
One application per student / scheme / year

Sanction ≤ scheme amount

Verification before approval

Disbursement ≤ sanctioned amount

Valid bank details before disbursement
```

Sample Data

Illustrative records verify completeness across the lifecycle.

APPLICATION	STUDENT	SCHEME	STATUS	AMOUNT	NEXT STEP
A-1042	Meera S.	Merit Grant	Verified	₹45,000	Approval
A-1043	Arjun K.	Need Aid	Pending	₹30,000	Documents
A-1044	Nila R.	Research	Approved	₹80,000	Sanction
A-1045	Dev P.	Merit Grant	Disbursed	₹45,000	Renewal

Also populated: eligibility criteria, documents, verifications, bank details, sanctions, disbursements and renewals.

SQL Queries, Joins & Views

Operational queries become reusable reporting views.

QUERY TOOLKIT

```
SELECT s.Name, sc.Scheme_Name, a.Status
FROM APPLICATION a
INNER JOIN STUDENT s ON s.Student_ID = a.Student_ID
LEFT JOIN SCHEME sc ON sc.Scheme_ID = a.Scheme_ID
WHERE a.Status = 'Pending';
```

SELECT • JOIN • SUM / COUNT • GROUP BY • SUBQUERY

- Pending applications report
- Eligible students list
- Approval status tracking
- Sanction + disbursement summaries
- Renewal monitoring
- Scheme statistics dashboard

Review 2 Implementation Progress

The database foundation is complete and ready for validation.

- ✓ ER model finalized
- ✓ Schema normalized to 3NF
- ✓ Dependencies identified
- ✓ Data dictionary completed
- ✓ DDL + constraints implemented
- ✓ Sample data populated
- ✓ Queries + views developed
- ✓ Prototype database created

NEXT Testing → User feedback → Refinement